

Inter-AS Routing: Deep Dive into BGP

Attributes, NEXT-HOP, and Hot Potato Routing

Bernal Manzanilla

Concordia University of Edmonton
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 - eBGP vs. iBGP
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The Border Gateway Protocol (BGP)

While OSPF handles routing inside an Autonomous System (AS), BGP is the protocol that glues the thousands of ISPs in the Internet together.

- **Reachability:** BGP provides each AS a means to obtain subnet reachability information from neighboring ASs, and to propagate that information to all routers internal to the AS.
- **eBGP (External BGP):** The BGP sessions that span two different ASs. This is how a gateway router learns which destinations are reachable through a neighboring ISP.
- **iBGP (Internal BGP):** The BGP sessions between routers within the *same* AS. This is how the gateway router distributes the external reachability information to all internal routers.

BGP Route Attributes: AS-PATH

In BGP, an autonomous system does not just advertise a destination; it advertises a *route* to that destination. A route consists of the subnet prefix and its BGP attributes.

- **The AS-PATH:** This attribute contains the list of Autonomous Systems through which the advertisement has passed.
- **How it works:** When a prefix is passed to an AS, the AS adds its own AS number (ASN) to the AS-PATH.
- **Loop Prevention:** BGP routers use the AS-PATH to detect and prevent looping advertisements. If a router sees its own AS number in the AS-PATH, it rejects the advertisement.

BGP Route Attributes: NEXT-HOP

The NEXT-HOP attribute is the critical link that bridges inter-AS routing (BGP) and intra-AS routing (OSPF).

- **Definition:** The NEXT-HOP is the IP address of the router interface that begins the AS-PATH. It is the IP address of the gateway router in the adjacent AS.
- **The Problem:** An internal router (using iBGP) knows it needs to use a specific AS-PATH to reach a destination, but how does it physically get the packet to that external gateway?
- **The Solution:** The internal router uses its intra-AS routing protocol (like OSPF) to find the least-cost path from itself to the IP address specified in the NEXT-HOP attribute.

BGP Route Selection

If a router learns about multiple routes to the same destination, how does it choose which one to use? It applies elimination rules in this order:

- 1 **Local Preference:** Routes are assigned a local preference value by the network administrator based on policy (e.g., "always prefer the path through ISP A over ISP B"). Highest preference wins.
- 2 **Shortest AS-PATH:** If preferences are equal, select the route with the fewest number of AS hops.
- 3 **Hot Potato Routing:** If AS-PATH lengths are equal, select the route with the closest NEXT-HOP router.
- 4 **BGP Identifiers:** If all else is equal, use BGP router identifiers to break the tie.

Hot Potato Routing

Hot potato routing is a selfish routing algorithm.

- **The Concept:** Think of the packet as a burning hot potato. The router wants to get the packet out of its own Autonomous System as quickly as possible to minimize its own costs.
- **The Mechanism:** The router looks at the multiple NEXT-HOP IP addresses available for the destination. It then consults OSPF to find the intra-AS cost to reach each of those gateways.
- **The Decision:** It chooses the gateway that has the smallest intra-AS cost, completely ignoring what happens to the packet outside of its own AS (the inter-AS costs).